

# TECHSHIP INTERNSHIP

## Day 3- Functions,Arrays,Objects & ES6 Concepts

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During Day 3 of the internship, I learned how to work with functions, arrays, objects, and modern JavaScript ES6 features. I gained a better understanding of how reusable functions help organize code and how arrays and objects can be used to store and manage structured data efficiently. I also explored array methods such as `forEach()`, `filter()`, `map()`, and `reduce()`, which are useful for processing and manipulating data in real-world applications.

The ES6 feature I found most useful was the spread operator because it allows objects and arrays to be copied and modified easily while keeping the original data unchanged. I also found template literals very helpful for generating clean and readable reports.

The main challenge I faced was understanding the `reduce()` method, particularly how its parameters work and the purpose of the initial value. While implementing the Student Management System assignment, I initially found it difficult to calculate averages and identify the top performer using array methods. However, after practicing with examples and debugging the code, I became more confident in using these concepts. Overall, this session improved my understanding of structured JavaScript programming, strengthened my problem-solving skills, and gave me hands-on experience in developing programs that resemble real-world data management systems.

The screenshot shows the Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows a folder structure with 'Day1', 'Day2', and 'Day3'. Under 'Day3', there is an 'Exercise' folder containing 'Ex1.js' and a 'README.md' file. The main editor window displays the code in 'Ex1.js', which defines four functions: 'add', 'subtract', 'multiply', and 'divide'. Each function takes two arguments, 'a' and 'b', and returns the result of the corresponding operation. Below the function definitions, there are four console.log statements that call each function with specific values: add(50, 100), subtract(12, 6), multiply(0, 3), and divide(2, 4). The bottom panel shows the 'TERMINAL' tab with the command 'node Ex1.js' executed, resulting in the following output: 'Addition: 150', 'Subtraction: 6', 'Multiplication: 0', and 'Division: 0.5'.

```
1 function add(a, b) {
2   return a + b;
3 }
4
5 function subtract(a, b) {
6   return a - b;
7 }
8
9 function multiply(a, b) {
10  return a * b;
11 }
12
13 function divide(a, b) {
14  return a / b;
15 }
16
17 console.log("Addition:", add(50, 100));
18 console.log("Subtraction:", subtract(12, 6));
19 console.log("Multiplication:", multiply(0, 3));
20 console.log("Division:", divide(2, 4));
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Exercise>node Ex1.js  
Addition: 150  
Subtraction: 6  
Multiplication: 0  
Division: 0.5  
D:\TECHSHIP\Day3\Exercise>

Ex1:Function Based Calculator

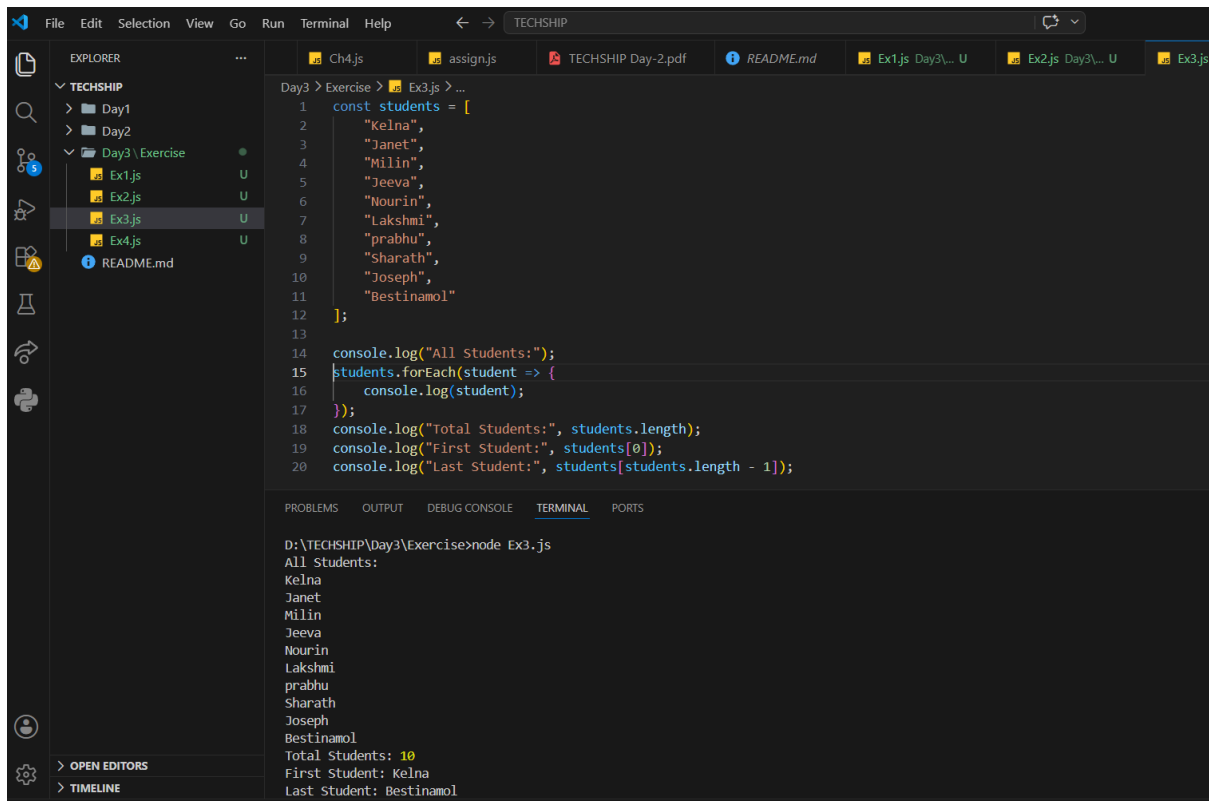
The screenshot shows the Visual Studio Code editor with the same 'TECHSHIP' project. The Explorer sidebar now shows 'Ex2.js' selected under the 'Day3' folder. The main editor window displays the code in 'Ex2.js', which defines a function 'greetStudent' that takes 'name' and 'course' as arguments. The function uses two console.log statements to output a welcome message and the course name. Below the function definition, there is a call to 'greetStudent' with the arguments 'Kelna' and 'MBA'. The bottom panel shows the 'TERMINAL' tab with the command 'node Ex2.js' executed, resulting in the following output: 'Welcome Kelna!' and 'You are enrolled in MBA.'.

```
1 function greetStudent(name, course) {
2   console.log(`Welcome ${name}!`);
3   console.log(`You are enrolled in ${course}.`);
4 }
5
6 greetStudent("Kelna", "MBA");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Exercise>node Ex2.js  
Welcome Kelna!  
You are enrolled in MBA.  
D:\TECHSHIP\Day3\Exercise>

Ex2:Student Greeting Function



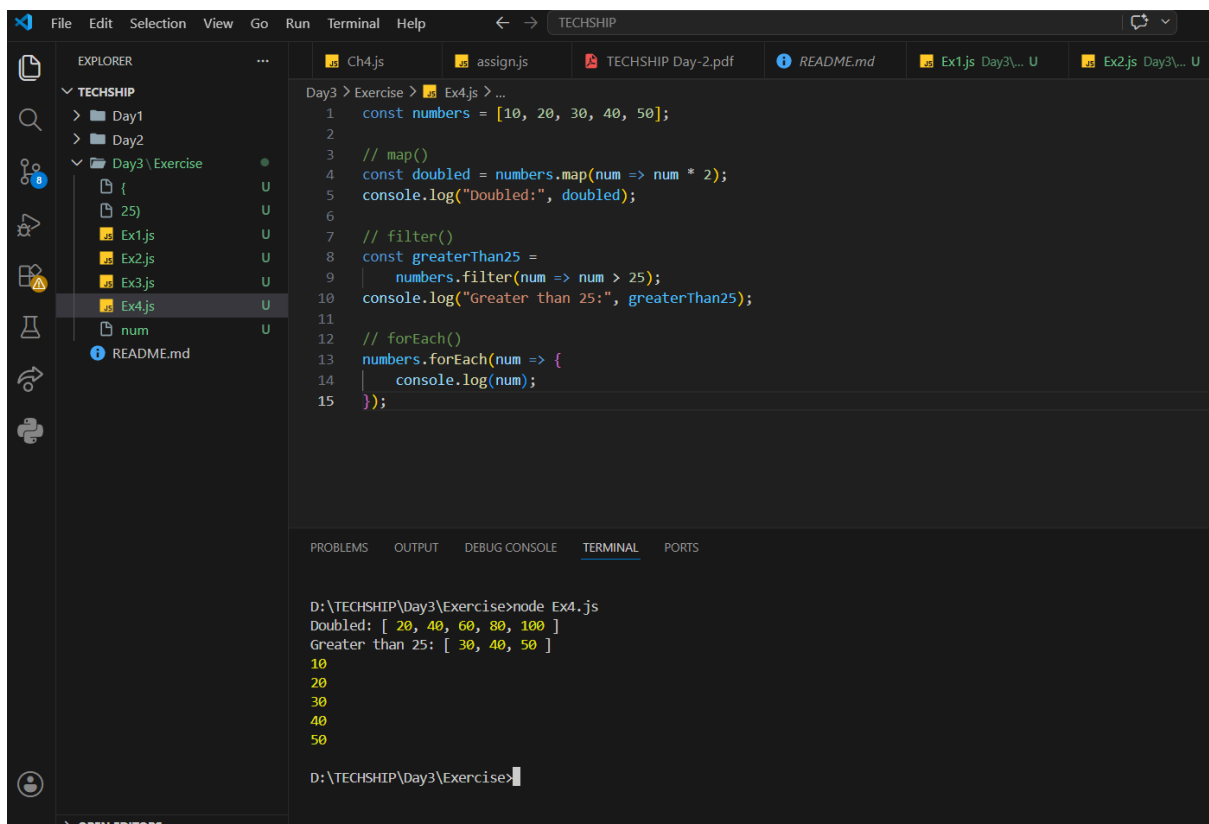
The screenshot shows the VS Code editor with a file named `Ex3.js` open. The file contains a JavaScript array of student names and a series of `console.log` statements to display the array and its details. The terminal output shows the execution of `node Ex3.js`, displaying the array of names, the total number of students (10), the first student (Kelna), and the last student (Bestinamol).

```
Day3 > Exercise > Ex3.js > ...
1  const students = [
2      "Kelna",
3      "Janet",
4      "Milin",
5      "Jeeva",
6      "Nourin",
7      "Lakshmi",
8      "prabhu",
9      "Sharath",
10     "Joseph",
11     "Bestinamol"
12 ];
13
14 console.log("All Students:");
15 students.forEach(student => {
16     console.log(student);
17 });
18 console.log("Total Students:", students.length);
19 console.log("First Student:", students[0]);
20 console.log("Last Student:", students[students.length - 1]);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Exercise>node Ex3.js  
All Students:  
Kelna  
Janet  
Milin  
Jeeva  
Nourin  
Lakshmi  
prabhu  
Sharath  
Joseph  
Bestinamol  
Total Students: 10  
First Student: Kelna  
Last Student: Bestinamol

## Ex3:Working with Arrays



The screenshot shows the VS Code editor with a file named `Ex4.js` open. The file contains a JavaScript array of numbers and a series of `console.log` statements to display the array and its details. The terminal output shows the execution of `node Ex4.js`, displaying the array of numbers, the doubled array, the array of numbers greater than 25, and the array of numbers less than or equal to 25.

```
Day3 > Exercise > Ex4.js > ...
1  const numbers = [10, 20, 30, 40, 50];
2
3  // map()
4  const doubled = numbers.map(num => num * 2);
5  console.log("Doubled:", doubled);
6
7  // filter()
8  const greaterThan25 =
9      numbers.filter(num => num > 25);
10 console.log("Greater than 25:", greaterThan25);
11
12 // forEach()
13 numbers.forEach(num => {
14     console.log(num);
15 });
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Exercise>node Ex4.js  
Doubled: [ 20, 40, 60, 80, 100 ]  
Greater than 25: [ 30, 40, 50 ]  
10  
20  
30  
40  
50  
D:\TECHSHIP\Day3\Exercise>

## Ex4:Array Methods

The screenshot shows the Visual Studio Code interface. The Explorer panel on the left displays a project structure with folders 'Day1' and 'Day2', and a folder 'Day3' containing an 'Exercise' subfolder. The 'Exercise' folder contains five JavaScript files: 'Ex1.js', 'Ex2.js', 'Ex3.js', 'Ex4.js', and 'Ex5.js'. The 'Ex5.js' file is selected and its content is displayed in the main editor. The code defines a 'student' object with properties: 'name' (Kelna), 'age' (21), 'college' (AISAT College), 'course' (BTech), and 'skills' (HTML, CSS, JavaScript, Flutter). The code also includes five console.log statements to log each property. The Terminal panel at the bottom shows the command 'node Ex5.js' being executed, resulting in the following output: 'Kelna', '21', 'AISAT College', 'BTech', and an array of skills: ['HTML', 'CSS', 'JavaScript', 'Flutter'].

```
1 const student = {
2   name: "Kelna",
3   age: 21,
4   college: "AISAT College",
5   course: "BTech",
6   skills: ["HTML", "CSS", "JavaScript", "Flutter"]
7 };
8
9 console.log(student.name);
10 console.log(student.age);
11 console.log(student.college);
12 console.log(student.course);
13 console.log(student.skills);
```

D:\TECHSHIP\Day3\Exercise>node Ex5.js  
Kelna  
21  
AISAT College  
BTech  
[ 'HTML', 'CSS', 'JavaScript', 'Flutter' ]  
D:\TECHSHIP\Day3\Exercise>

Ex5:Student Object

The screenshot shows the Visual Studio Code interface. The Explorer panel on the left displays a project structure with folders 'Day1' and 'Day2', and a folder 'Day3' containing an 'Exercise' subfolder. The 'Exercise' folder contains five JavaScript files: 'Ex1.js', 'Ex2.js', 'Ex3.js', 'Ex4.js', and 'Ex5.js'. The 'Ex5.js' file is selected and its content is displayed in the main editor. The code defines an 'employee' object with properties: 'name' (Kelna), 'age' (21), 'college' (AISAT College), 'course' (BTech), and 'skills' (HTML, CSS, JavaScript, Flutter). The code also includes five console.log statements to log each property. The Terminal panel at the bottom shows the command 'node Ex5.js' being executed, resulting in the following output: 'Kelna', '21', 'AISAT College', 'BTech', and an array of skills: ['HTML', 'CSS', 'JavaScript', 'Flutter'].

```
1 const employee = {
2   name: "Kelna",
3   age: 21,
4   college: "AISAT College",
5   course: "BTech",
6   skills: ["HTML", "CSS", "JavaScript", "Flutter"]
7 };
8
9 console.log(employee.name);
10 console.log(employee.age);
11 console.log(employee.college);
12 console.log(employee.course);
13 console.log(employee.skills);
```

D:\TECHSHIP\Day3\Exercise>node Ex5.js  
Kelna  
21  
AISAT College  
BTech  
[ 'HTML', 'CSS', 'JavaScript', 'Flutter' ]  
D:\TECHSHIP\Day3\Exercise>

Ex6:Employee Object

The screenshot shows a VS Code editor with a file explorer on the left displaying a project structure for 'TECHSHIP'. The file 'Ex6.js' is selected under 'Day3 > Exercise'. The code in 'Ex6.js' defines an employee object with the following properties: employeeId: 102, name: 'Aadhithya', department: 'HR', and salary: 30000. It then logs the object before and after updating the salary to 40000. The terminal at the bottom shows the command 'D:\TECHSHIP\Day3\Exercise>node Ex6.js' and its output, which matches the code's console.log statements.

```
1 const employee = {
2   employeeId: 102,
3   name: "Aadhithya",
4   department: "HR",
5   salary: 30000
6 };
7
8 console.log("Before Update:");
9 console.log(employee);
10
11 // Update salary
12 employee.salary = 40000;
13
14 console.log("After Update:");
15 console.log(employee);
```

```
D:\TECHSHIP\Day3\Exercise>node Ex6.js
Before Update:
{ employeeId: 102, name: 'Aadhithya', department: 'HR', salary: 30000 }
After Update:
{ employeeId: 102, name: 'Aadhithya', department: 'HR', salary: 40000 }
```

## Ex7:ES6 Practise

## Challenge Problems

The screenshot shows a VS Code editor with a file explorer on the left displaying a project structure for 'TECHSHIP'. The file 'Ex7.js' is selected under 'Day3 > Exercise'. The code in 'Ex7.js' demonstrates several ES6 features: a const variable for a student object, a template literal for a log statement, destructuring to extract name and course, and the spread operator to create a new student object with an additional college property. The terminal at the bottom shows the command 'D:\TECHSHIP\Day3\Exercise>node Ex7.js' and its output, which matches the code's console.log statements.

```
1 const student = {
2   name: 'Kelna',
3   age: 21,
4   course: 'BTech CSE'
5 };
6 //Template Literal
7 console.log(
8   `Hi, I am ${student.name}.
9   I am studying ${student.course}.`
10 );
11 //Destructuring
12 const { name, course } = student;
13 console.log(name);
14 console.log(course);
15 //Spread Operator
16 const newStudent = {
17   ...student,
18   college: "XYZ College"
19 };
20 console.log(newStudent);
```

```
D:\TECHSHIP\Day3\Exercise>node Ex7.js
Hi, I am Kelna.
I am studying BTech CSE.
Kelna
BTech CSE
{ name: 'Kelna', course: 'BTech CSE', age: 21, college: 'XYZ College' }
```

## 1.Student Marks Analyser

```
File Edit Selection View Go Run Terminal Help
TECHSHIP

EXPLORER
TECHSHIP
├── Day1
├── Day2
│   ├── Challenges
│   └── Exercise
│       ├── assign.js
│       └── TECHSHIP Day-2.pdf
├── Day3
│   ├── Challenge
│   │   ├── Ch1.js
│   │   └── Exercise
│   │       ├── Ex1.js
│   │       ├── Ex2.js
│   │       ├── Ex3.js
│   │       ├── Ex4.js
│   │       ├── Ex5.js
│   │       ├── Ex6.js
│   │       └── Ex7.js
│   └── README.md
└── ...

Day3 > Challenge > Ch1.js > marks
1 const marks = [90, 40, 86, 65, 78];
2
3 // Total
4 const total = marks.reduce((sum, mark) => sum + mark, 0);
5
6 // Highest
7 const highest = Math.max(...marks);
8
9 // Lowest
10 const lowest = Math.min(...marks);
11
12 // Average
13 const average = total / marks.length;
14
15 console.log("Total Marks:", total);
16 console.log("Highest Mark:", highest);
17 console.log("Lowest Mark:", lowest);
18 console.log("Average Mark:", average);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Challenge>node Ch1.js
Total Marks: 359
Highest Mark: 90
Lowest Mark: 40
Average Mark: 71.8

D:\TECHSHIP\Day3\Challenge>
```

## 2.Product Filter System

```
File Edit Selection View Go Run Terminal Help
TECHSHIP

EXPLORER
TECHSHIP
├── Day1
├── Day2
│   ├── Challenges
│   └── Exercise
│       ├── assign.js
│       └── TECHSHIP Day-2.pdf
├── Day3
│   ├── Challenge
│   │   ├── Ch1.js
│   │   ├── Ch2.js
│   │   └── Exercise
│   │       ├── Ex1.js
│   │       ├── Ex2.js
│   │       ├── Ex3.js
│   │       ├── Ex4.js
│   │       ├── Ex5.js
│   │       ├── Ex6.js
│   │       └── Ex7.js
│   └── README.md
└── ...

Day3 > Challenge > Ch2.js > expensiveProducts > products.filter() callback
1 const products = [
2   { name: "Mouse", price: 700 },
3   { name: "Keyboard", price: 5000 },
4   { name: "Monitor", price: 20000 },
5   { name: "Speaker", price: 5000 }
6 ];
7
8 const expensiveProducts =
9   products.filter(product => product.price > 1000);
10
11 console.log(expensiveProducts);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Challenge>node Ch2.js
[
  { name: 'Keyboard', price: 5000 },
  { name: 'Monitor', price: 20000 },
  { name: 'Speaker', price: 5000 }
]
```

## 3.Employee Directory

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying a project structure. The main editor area shows the code for `Ch3.js` in the `Day3 > Challenge` directory. The code defines an array of employee objects and uses `forEach` to log their names, and `filter` to log employees in the IT department. The terminal at the bottom shows the command `node Ch3.js` being executed, resulting in the names of the employees and a filtered array of IT department employees.

```
Day3 > Challenge > Ch3.js > ...
1  const employees = [
2    { id: 1,
3      name: "Renu",
4      department: "IT"},
5    { id: 2,
6      name: "Jithu",
7      department: "HR"},
8    {id: 3,
9      name: "Geetha",
10     department: "IT"}];
11 // All employee names
12 employees.forEach(employee => {
13   console.log(employee.name);
14 });
15 // IT Department employees
16 const itEmployees =
17   employees.filter(
18     employee => employee.department === "IT");
19 console.log(itEmployees);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
D:\TECHSHIP\Day3\Challenge>node Ch3.js
Renu
Jithu
Geetha
[
  { id: 1, name: 'Renu', department: 'IT' },
  { id: 3, name: 'Geetha', department: 'IT' }
]

D:\TECHSHIP\Day3\Challenge>
```

#### 4.Course Enrollment System

# Assignment

```
Ex6.js Day3... U  Ex7.js Day3... U  Ch1.js ...\\Challenge U  Ch2.js ...\\Challenge U  Ch3.js ...\\Challenge U  Ch4.js ...\\Challenge U
Day3 > assign.js > studentAverages > students.map() callback
1  const students = [{id: 100,
2      name: "Evelin",
3      age: 20,
4      course: "Cardiology Technician",
5      marks: [75, 80, 70, 85, 90]}, {
6      id: 101,
7      name: "Ann",
8      age: 21,
9      course: "BTech EC",
10     marks: [79, 82, 75, 88, 80]}, {
11     id: 102,
12     name: "Janet",
13     age: 21,
14     course: "BTech CS",
15     marks: [90, 95, 92, 88, 91]}, {
16     id: 103,
17     name: "Keina",
18     age: 22,
19     course: "BTech AI/ML",
20     marks: [65, 70, 68, 72, 66]},
21     { id: 104,
22         name: "Divya",
23         age: 21,
24         course: "Bsc Nurse",
25         marks: [78, 80, 75, 82, 79] }];
26     // Calculate average for each student
27     const studentAverages = students.map(student => {
28     const total = student.marks.reduce(
29         (sum, mark) => sum + mark,
30         0);
31     const average = total / student.marks.length;
32     return {
33         ...student,
34         average: average};});
35     // Students scoring above 75%
36     const topStudents = studentAverages.filter(
37         student => student.average > 75);
38     // Find top performer
39     const topper = studentAverages.reduce(
40         (highest, student) =>
41             student.average > highest.average
42             ? student
43             : highest);
44     console.log("=====");
45     console.log("STUDENT MANAGEMENT REPORT");
46     console.log("=====");
47     // Display all students with report
48     studentAverages.forEach(student => {let status;
49     if (student.average >= 80) {
50         status = "Excellent";
51     } else if (student.average >= 70) {
52         status = "Good";
53     } else {
54         status = "Needs Improvement";}
55     console.log(`ID: ${student.id}
56     Name: ${student.name}
57     Course: ${student.course}
58     Average Marks: ${student.average.toFixed(2)}
59     Status: ${status}`);});
60     // Students above 75%
61     console.log("=====");
62     console.log("Students Scoring Above 75%:");
63     topStudents.forEach(student => {console.log(`${student.name} - ${student.average.toFixed(2)} `)});});
64     // Top performer
65     console.log("=====");
66     console.log(`Top Performer: ${topper.name}`);
67     console.log(`Highest Average: ${topper.average.toFixed(2)} `);
68     console.log("=====");
```

## Student Management System



